

❖ Ch3_No.5

For the two-energy levels system, the spontaneous emission lifetime of energy level E_2 is τ_{s2} , and the nonradiative recombination lifetime is τ_{nr2} . If the carrier density at energy level E_2 at $t=0$ is $n_2(0)$, the volume of the gain material is V , the frequency of the spontaneous emission is ν , then please calculate:

- 1) The optical power of the spontaneous emission change with time t .
- 2) The photon number of the spontaneous emission from the carriers in energy level E_2 during the decay process.
- 3) The ratio η_z of the spontaneous emission photon number with the initial carrier number at energy level E_2 . η_z is known as the quantum yield.

$$\text{total lifetime: } \frac{1}{\tau_z} = \frac{1}{\tau_{s2}} + \frac{1}{\tau_{nr2}}$$

$$A_z = A_{s2} + A_{nr2}$$

$$n_2(t) = n_2(0) \cdot \exp\left(-\frac{t}{\tau_z}\right)$$

$$\Rightarrow N_2(t) = V \cdot n_2(t)$$

$$1) P_{sp}(t) = h\nu \cdot \frac{N_2(t)}{\tau_{s2}} = \frac{h\nu \cdot V \cdot n_2(0)}{\tau_{s2}} \cdot \exp\left(-\frac{t}{\tau_z}\right)$$

$$\Rightarrow P_{sp}(t) = \frac{h\nu \cdot V}{\tau_{s2}} \cdot n_2(0) \cdot \exp\left(-\frac{\tau_{s2} + \tau_{nr2}}{\tau_{s2} \cdot \tau_{nr2}} \cdot t\right)$$

$$2) N_{ph} = \int_0^\infty \frac{N_2(t)}{\tau_{s2}} dt = \int_0^\infty \frac{V \cdot n_2(0)}{\tau_{s2}} \cdot \exp\left(-\frac{t}{\tau_z}\right) dt$$

$$\Rightarrow N_{ph} = \frac{V \cdot n_2(0)}{\tau_{s2}} \left[-\tau_z \cdot \exp\left(-\frac{t}{\tau_z}\right) \right]_0^\infty$$

$$= \frac{V \cdot n_2(0)}{\tau_{s2}} \cdot \frac{\tau_{s2} \cdot \tau_{nr2}}{\tau_{s2} + \tau_{nr2}}$$

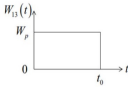
$$= V \cdot n_2(0) \cdot \frac{\tau_{nr2}}{\tau_{s2} + \tau_{nr2}}$$

$$3) N_2(0) = V \cdot n_2(0)$$

$$\eta_z = \frac{N_{ph}}{N_2(0)} = \frac{\tau_{nr2}}{\tau_{s2} + \tau_{nr2}}$$

❖ Ch3_No.8

Assuming that the ruby with a carrier density of n is excited by a rectangular pulse, the pumping transition probability can be expressed as

$$W_{13}(t) = \begin{cases} W_p & 0 < t \leq t_0 \\ 0 & t > t_0 \end{cases}$$


$$n_3 \rightarrow 0$$

$$\approx$$

Please determine the upper carrier density $n_3(t)$, and then draw the corresponding curve.

(Note: S_{32} is very big and therefore $n_3 \approx 0$, $dn_3/dt = 0$. The stimulated transition between the up and down energy levels can be ignored without optical cavity.)

$$n_3 \approx 0, \frac{dn_3}{dt} = 0 \Rightarrow n = n_1 + n_2 + n_3 \rightarrow n_1 = n - n_2$$

$$\frac{dn_3}{dt} = W_{13} \cdot n_1 - S_{32} \cdot n_3 = 0 \rightarrow S_{32} \cdot n_3 = W_{13} \cdot n_1$$

$$\frac{dn_2}{dt} = S_{32} \cdot n_3 - \frac{n_2}{\tau} = W_{13} \cdot n_1 - \frac{n_2}{\tau}$$

$$\Rightarrow \frac{dn_2}{dt} = W_{13}(t) \cdot (n - n_2) - \frac{n_2}{\tau}$$

$$\textcircled{1} \quad 0 < t < t_0, \quad W_{13}(t) = W_p, \quad n_2(0) = 0$$

$$\frac{dn_2}{dt} = W_p n - (W_p + \frac{1}{\tau}) n_2$$

$$n_2(t) = \frac{W_p \cdot n}{W_p + \frac{1}{\tau}} \cdot (1 - \exp(-(W_p + \frac{1}{\tau})t))$$

$\hookrightarrow t_0 \rightarrow \text{peak}$

$$\textcircled{2} \quad t > t_0$$

$$\frac{dn_2}{dt} = -\frac{n_2}{\tau}$$

$$\Rightarrow n_2(t) = n_2(t_0) \cdot \exp\left(-\frac{t-t_0}{\tau}\right) = \frac{W_p \cdot n}{W_p + \frac{1}{\tau}} \left(1 - \exp(-(W_p + \frac{1}{\tau})t_0)\right) \cdot \exp\left(-\frac{t-t_0}{\tau}\right)$$

curve:

